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[Screenshot # 1: Show the completed RDS configuration page before clicking Create]

RDS configuration page

RDS

> Create database

Create database [Info](#)

Choose a database creation method

☒ Standard create
You set all of the configuration options, including ones for availability, security, backups, and maintenance.

☐ Easy create
Use recommended best-practice configurations. Some configuration options can be changed after the database is created.

Engine options

Engine type [Info](#)

☐ Aurora (MySQL Compatible)


☒ MySQL


☐ MariaDB


☐ Aurora (PostgreSQL Compatible)


☐ PostgreSQL


☐ Oracle


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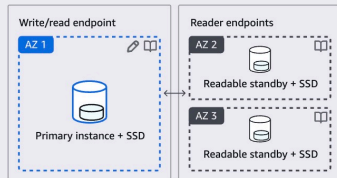
Change into Free tier

- ☐ **Production**
Use defaults for high availability and fast, consistent performance.

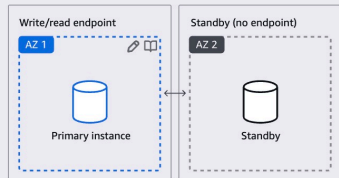
- **Free tier**
Use RDS Free Tier to develop new applications, test existing applications, or gain hands-on experience with Amazon RDS. [Info](#)

Deployment options [Info](#)

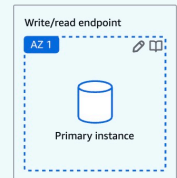
- Multi-AZ DB cluster deployment (3 instances)
Creates a primary DB instance with two readable standbys in separate Availability Zones. This setup provides:
 - 99.95% uptime
 - Redundancy across Availability Zones
 - Increased read capacity
 - Reduced write latency



- Multi-AZ DB instance deployment (2 instances)
 - Creates a primary DB instance with a non-readable standby instance in a separate Availability Zone. This setup provides:
 - 99.95% uptime
 - Redundancy across Availability Zones



- **Single-AZ DB instance deployment (1 instance)**
Creates a single DB instance without standby instances. This setup provides:
 - 99.5% uptime
 - No data redundancy



☰ [RDS](#) > Create database

DB instance identifier [Info](#)

Type a name for your DB instance. The name must be unique across all DB instances owned by your AWS account in the current AWS Region.

The DB instance identifier is case-insensitive, but is stored as all lowercase (as in "mydbinstance"). Constraints: 1 to 63 alphanumeric characters or hyphens. First character must be a letter. Can't contain two consecutive hyphens. Can't end with a hyphen.

▼ Credentials Settings

Master username [Info](#)

Type a login ID for the master user of your DB instance.

1 to 16 alphanumeric characters. The first character must be a letter.

Credentials management

You can use AWS Secrets Manager or manage your master user credentials.

○

Managed in AWS Secrets Manager - *most secure*
RDS generates a password for you and manages it throughout its lifecycle using AWS Secrets Manager.

☒ Self managed

Create your own password or have RDS create a password that you manage.

☐ Auto generate password

Amazon RDS can generate a password for you, or you can specify your own password.

Master password | Info

●●●●●●●●●●

Password strength **Very strong**

Minimum constraints: At least 8 printable ASCII characters. Can't contain any of the following symbols: / ' " @

Confirm master password [Info](#)

●●●●●●●●●●

▼ Hide filters

☐ Show instance classes that support Amazon RDS Optimized Writes

[Info](#)

Amazon RDS Optimized Writes improves write throughput by up to 2x at no additional cost.

☐ Include previous generation classes

☐ Standard classes (includes m classes)

☐ Memory optimized classes (includes r and x classes)

☒ Burstable classes (includes t classes)

db.t3.micro

2 vCPUs 1 GiB RAM Network: Up to 2,085 Mbps

Storage

Storage type [Info](#)

Provisioned IOPS SSD (io2) storage volumes are now available.

General Purpose SSD (gp2)

Baseline performance determined by volume size

Allocated storage [Info](#)

20

GiB

Allocated storage value must be 20 GiB to 6,144 GiB

► Additional storage configuration

public access: Yes, this is important for later connecting database

DB subnet group [Info](#)

Choose the DB subnet group. The DB subnet group defines which subnets and IP ranges the DB instance can use in the VPC that you selected.

default ▼

Public access [Info](#)

☒ Yes

RDS assigns a public IP address to the database. Amazon EC2 instances and other resources outside of the VPC can connect to your database. Resources inside the VPC can also connect to the database. Choose one or more VPC security groups that specify which resources can connect to the database.

☐ No

RDS doesn't assign a public IP address to the database. Only Amazon EC2 instances and other resources inside the VPC can connect to your database. Choose one or more VPC security groups that specify which resources can connect to the database.

VPC security group (firewall) [Info](#)

Choose one or more VPC security groups to allow access to your database. Make sure that the security group rules allow the appropriate incoming traffic.

☒ Choose existing

Choose existing VPC security groups

☐ Create new

Create new VPC security group

Existing VPC security groups

Choose one or more options ▼

default ✕

Availability Zone [Info](#)

No preference ▼

RDS Proxy

RDS Proxy is a fully managed, highly available database proxy that improves application scalability, resiliency, and security.

☐ Create an RDS Proxy [Info](#)

RDS automatically creates an IAM role and a Secrets Manager secret for the proxy. RDS Proxy has additional costs. For more information, see [Amazon RDS Proxy pricing](#) [\[2\]](#).

▼ Additional configuration

Database port [Info](#)

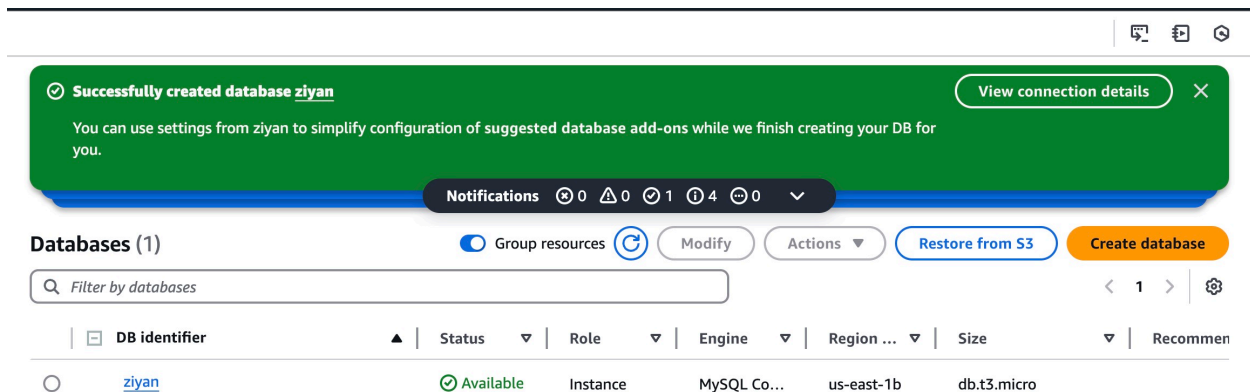
TCP/IP port that the database will use for application connections.

3306

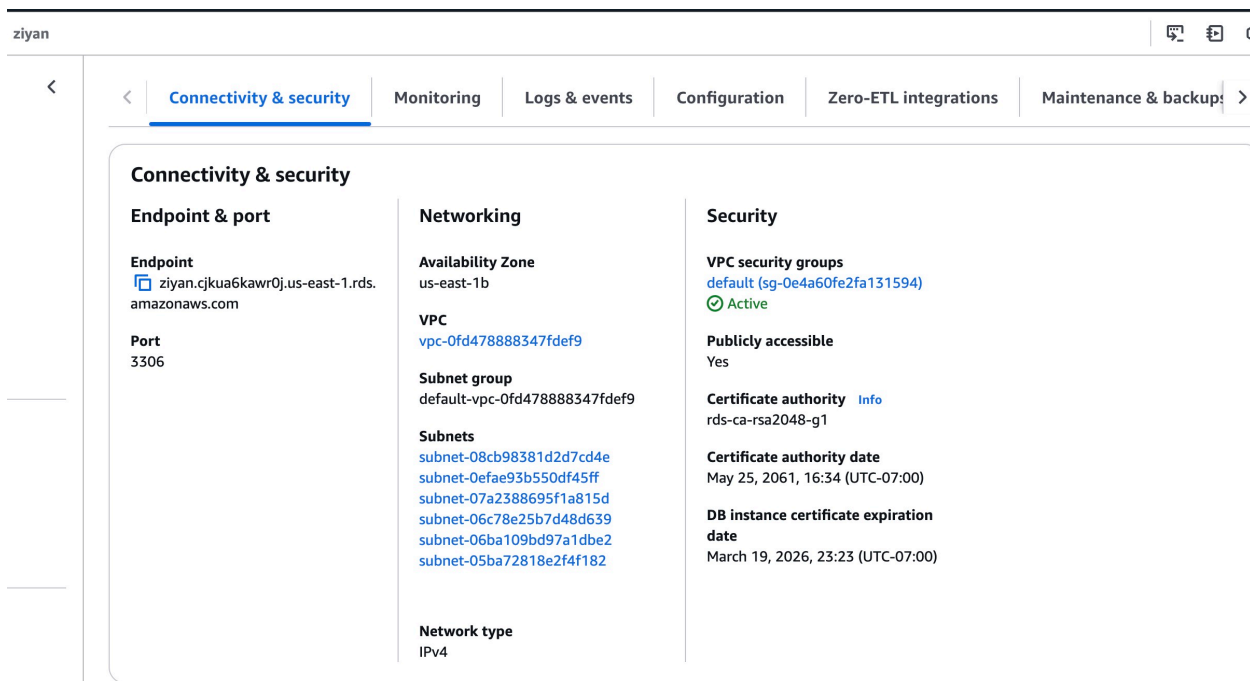
Disable Backup (uncheck).Disable Encryption (uncheck).Disable minor version upgrade(uncheck)

[Screenshot #2: Show the successful database creation and endpoint from the Connectivity & Security section.]

database created successfully



Endpoint show in the connectivity & security



[ziyan.cjkua6kawr0j.us-east-1.rds.amazonaws.com](#)

this endpoint need to be copied for later use

In Cloushell, connect to the RDS Database successfully:

```
~ $ mysql -h ziyan.cjkua6kawr0j.us-east-1.rds.amazonaws.com -P 3306 -u admin -p
Enter password:
Welcome to the MariaDB monitor.  Commands end with ; or \g.
Your MySQL connection id is 28
Server version: 8.0.40 Source distribution

Copyright (c) 2000, 2018, Oracle, MariaDB Corporation Ab and others.

Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

MySQL [(none)]> 
```

[Screenshot #3: Show the successful table creation in the CloudShell terminal.]
run **SHOW TABLES;** and see the table has created successfully:

```
MySQL [(none)]> CREATE DATABASE student_records;
Query OK, 1 row affected (0.003 sec)

MySQL [(none)]> USE student_records;
Database changed
MySQL [student_records]> CREATE TABLE students (
    ->  student_id INT PRIMARY KEY,
    ->  name VARCHAR(100),
    ->  major VARCHAR(50),
    ->  enrollment_date DATE
    -> );
Query OK, 0 rows affected (0.030 sec)

MySQL [student_records]> SHOW TABLES;
+-----+
| Tables_in_student_records |
+-----+
| students                  |
+-----+
1 row in set (0.002 sec)

MySQL [student_records]> 
```

[Screenshot #4: Show the query results in the CloudShell terminal.]

insert data into the table and show query:

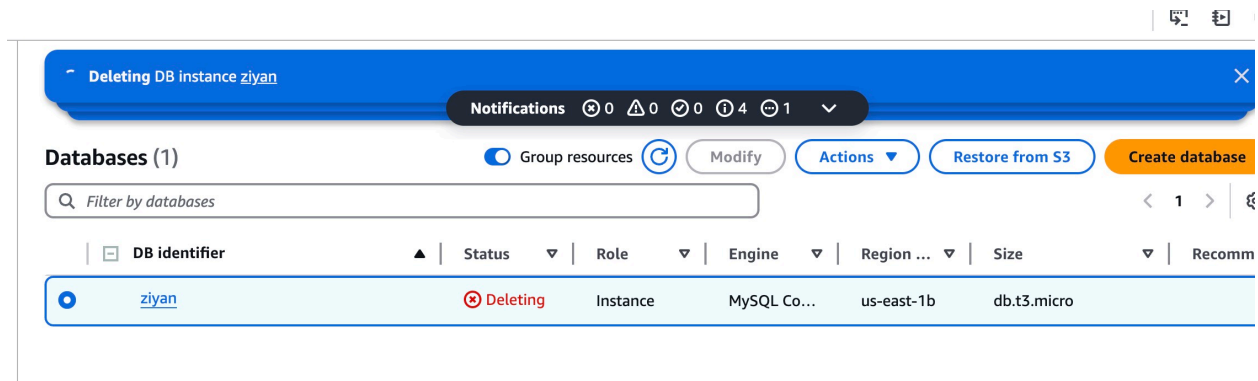
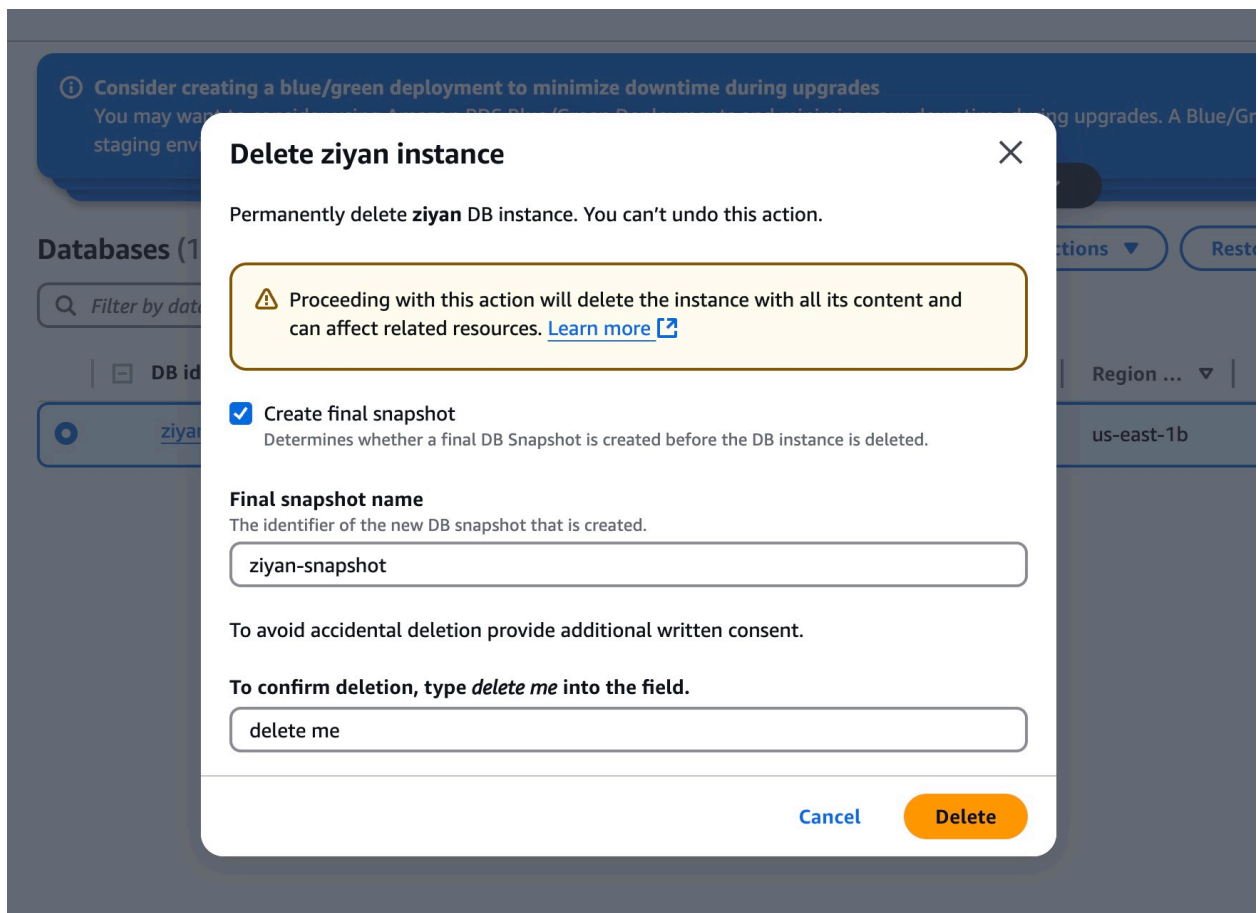
```
MySQL [student_records]> INSERT INTO students (student_id, name, major, enrollment_date)
-> VALUES
-> (1, 'Alice Johnson', 'Computer Science', '2023-08-21'),
-> (2, 'Bob Smith', 'Information Systems', '2024-01-15');
Query OK, 2 rows affected (0.003 sec)
Records: 2  Duplicates: 0  Warnings: 0

MySQL [student_records]> SELECT * FROM students;
+-----+-----+-----+-----+
| student_id | name       | major           | enrollment_date |
+-----+-----+-----+-----+
| 1          | Alice Johnson | Computer Science | 2023-08-21      |
| 2          | Bob Smith    | Information Systems | 2024-01-15      |
+-----+-----+-----+-----+
2 rows in set (0.002 sec)

MySQL [student_records]> 
```

[Screenshot #5: Show the successful deletion of the RDS database.]

confirm before delete:



Delete the RDS instance successfully, no instances are shown now:

Consider creating a blue/green deployment to minimize downtime during upgrades

You may want to consider using Amazon RDS Blue/Green Deployments and minimize your downtime during upgrades. A Blue/Green Deployment provides a staging environment for changes to production databases. [RDS User Guide](#) [Aurora User Guide](#)

Notifications

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0

Databases (0)

Group resources

Modify

Actions

Restore from S3

Create database

Filter by databases

<

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DB identifier		Status	Role	Engine	Region ...	Size		Recommen
No instances found								